Resume

Dr. Abhijit Datta

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Google Scholar Profile: https://scholar.google.com/citations?user=t0CryP4AAAAJ&hl=en, total citations 24.
Researchgate Profile: https://www.researchgate.net/profile/Abhijit-Datta-6?ev=hdr_xprf, total citations 26.

Permanent and Present Address
Nischintapur, Rampurhat, Birbhum, West Bengal – 731224.
Date of Birth: December 27, 1980



Summary of qualifications, skills and experiences

- Awarded a Ph.D in Mechanical Engineering from a **Central Government Autonomous University (AcSIR-CSIR)** in December-2025. Thesis work entitled "Optimization of process parameters in Friction stir welding of aluminium to steel" carried out at CSIR – Central Mechanical Engineering Research Institute, Durgapur.
- Acquired total 8 years of Academic and Administrative Experience (from 2012 to 2020) at different Undergraduate and Diploma Institutions (in chronological order): RK University - School of Engineering, Rajkot, Gujrat; Santiniketan Institute of Polytechnic, Bolpur; HSB Polytechnic, Siuri; Kanad Institute of Engineering & Management, Mankar; Dr. B.C. Roy Polytechnic, Durgapur in the level of Assistant Professor, Lecturer and HOD.
- Obtained total 5+ years of Industrial experience up to the level of Engineer (Production) at Gajra Differential Gears Ltd (A Gajra Group Co.), Dewas, MP working there from 2009 to 2012. Worked at Alliance-Infotech Pvt. Ltd. (Delhi) and Forbes Campbell Knitwear (Belgaum, Karnataka) for a cumulative period of two years.
- Gained total 5+ years research experiences as a research scholar in Ph.D, expanded technical skills in Friction Stir Welding, Dissimilar Metal Joining, Parametric Optimization, Design of Experiments and Metallographic & Microstructural Characterization.
- Have experience in teaching the following subjects: Manufacturing Processes, Machining and Machine Tools, Non-conventional Machining, Machine Design, Engineering Mechanics, Engineering Drawing, Fluid Mechanics, Operation Research and Industrial Management etc.
- Acquainted in Design and Analysis of Experiments (Software: Design Experts and Minitab), Metallographic characterizations (OM, SEM and EDX), AutoCAD and Microsoft Office.

Academic Qualification

Degree	Year	Specialization	University/Institution	% or CGPA
Doctor of Philosophy (Ph.D)	Dec-2025	Engineering Science (Mechanical Engineering - Manufacturing)	Academy of Scientific and Innovative Research / CSIR – Central Mechanical Engineering Research Institute (CMERI), Durgapur	CGPA (Cumulative Grade Point Average, in Course-work) – 8.11 (on a 10-point scale)
Master of Technology (M.Tech)	2015	Mechanical Engineering (Design and Production)	Maulana Abul Kalam Azad University / Dr. B. C. Roy Engg. College, Durgapur	DGPA (Degree Grade Point Average) – 9.45 (on a 10-point scale)
Master of Business Administration (MBA)	2006	Marketing	Karnatak University Dharwad / KLS Institute of Management Education & Research, Belgavi	69%
Bachelor of Engineering (BE)	2004	Mechanical Engineering	Visveswaraiah Technological University / K.L.E. Society's College of Engineering & Technology, Belgavi	70.66%
(10+2)th	1999	Science	West Bengal Council of Higher Secondary Education / Asansol Dhadka NCLV	75.4%
10th	1997	Standard	West Bengal Board of Secondary Education / Asansol Ram Krishna Mission	82.5%

Research Publications

Journal papers

1. **Datta, A**., Shrivastava, A., Mandal, N., Roy, H., Chakraborty, S.S. A comparative investigation of butt friction stir welding of aluminium alloys, AA 1100 and AA 7075, with AISI 304 stainless steel. Welding in the world (Springer), March 2023, https://doi.org/10.1007/s40194-023-01514-6**.**
2. **Datta, A**., Mandal, N., Mukherjee, M., Bapanapalle, C.O., Sarkar, S., Chakraborty, S.S. Parametric optimisation in friction stir butt welding of spring steel to commercially pure aluminium sheet. Journal of Materials Engineering and Performance (Springer), Ref. ID-JMEP-25-07-43278 **(Under review).**
3. Shrivastava, A., Changdar, A., **Datta, A.**, Dutta, S., Chakraborty, S.S. Parametric investigation and optimization in laser based directed energy deposition of tungsten carbide-cobalt. The Journal of Laser Applications, November 2023, https://doi.org/10.2351/7.0001179.
4. **Datta, A.**, Shrivastava, A., Chakraborty, S.S., Pal, S. K. (2022). Effect of work material combination and welding speed in dissimilar friction stir welding of aluminium to steel. Materials Today: Proceedings (Elsevier) (SCOPUS), September 2022, https://doi.org/10.1016/j.matpr.2022.05.558.
5. Aditya, S. K., **Datta,** A., Analysis of Friction Stir Welding of Aluminum Alloys and Optimization of Welding Parameters for Maximum Tensile Strength**,** International Journal of Engineering Research & Application, ISSN: 2248-9622, Vol. 5, Issue 5, (Part -5) May 2015, pp. 36-43, https://www.ijera.com/papers/Vol5_issue5/Part%20-%205/H505053643.pdf.

Book Chapters

1. **Datta, A**., Mandal, N., Chakraborty, S.S., Application of friction stir processing for generating novel engineering parts – A review, Springer nature book chapter: Advances in Materials, Manufacturing and Design, Lecture Notes in Mechanical Engineering, January 2025, Page no.- 487-500, https://doi.org/10.1007/978-981-97-6667-3_38.

Conference Papers

1. **Datta, A**., Mandal, N., Chakraborty, S.S., Variation of friction stir processing for generating novel engineering parts – A review, INCOM24: Proceedings of the 2nd International Conference on Mechanical Engineering Jadavpur University, Kolkata, India, January 5-6, 2024,Page no 353-356, **ISBN: 978-81-954806-6-1**, Paper Id. INCOM24- 308.
2. **Datta, A**., Chakraborty, S.S., Mandal, N., Weld quality assessment and monitoring - a review**,** Thirty-five National Convention of Emerging Technologies for Sustainable Manufacturing & National Conference, The Institution of Engineers (India) (IEI), Durgapur, October 30-31, 2021, Volume page no.- 93-98, **ISBN: 978-81-952159-9-7.**
3. **Datta A**., Chakraborty S.S., Butt friction stir welding of steel to aluminium - a review, Thirty-third National Convention of Metallurgical and Materials Engineers & National Conference, The Institution of Engineers (India) (IEI), Durgapur, Volume page no.- 93-98, January 17-18, 2020.
4. Chakraborty, S.S., **Datta, A**., Mandal, N., Shrivastava, A., Mukherjee, M., A brief review on reported novel applications of friction stir processing**,** Thirty-five National Convention of Emerging Technologies for Sustainable Manufacturing & National Conference, The Institution of Engineers (India) (IEI), Durgapur, October 30-31, 2021, Volume page no.- 93-98, **ISBN: 978-81-952159-9-7**.

Languages known: (Speak, Read and Write) Bengali, Hindi and English.

Awards & Achievements

1. Received **Gold Medal** for being first class first in M. Tech.
2. **Best Paper Award** – National Convention, The Institution of Engineers (India), 2021.